

Fall 2025 CPTS530 Homework 5
MATH 448-548
Numerical Analysis

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November 2025

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Problem 1

Problem Statement (Textbook 4.4.1):

Show that the norms $\|\mathbf{x}\|_\infty$, $\|\mathbf{x}\|_2$, and $\|\mathbf{x}\|_1$ satisfy the postulates (1), (2), and (3) for norms.

Norm Postulates [1, 2]:

On a vector space V , a **norm** is a function $\|\cdot\|$ from V to the set of nonnegative reals that obeys these three postulates:

$$\|x\| > 0 \text{ if } x \neq 0, x \in V \tag{1}$$

$$\|\lambda x\| = |\lambda| \|x\| \text{ if } \lambda \in \mathbb{R}, x \in V \tag{2}$$

$$\|x + y\| \leq \|x\| + \|y\| \text{ if } x, y \in V \quad (\text{triangle inequality}) \tag{3}$$

Solution:

Problem 2

Problem Statement (Textbook 4.4.2):

Show that $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ for all $\mathbf{x} \in \mathbb{R}^n$, and that equalities can occur, even for nonzero vectors.

Solution:

Problem 3

Problem Statement (Textbook 4.4.3):

(Continuation) Show that $\|\mathbf{x}\|_1 \leq n\|\mathbf{x}\|_\infty$ and $\|\mathbf{x}\|_2 \leq \sqrt{n}\|\mathbf{x}\|_\infty$ for $\mathbf{x} \in \mathbb{R}^n$.

Solution:

Problem 4

Problem Statement (Textbook 4.4.8):

Define

$$\|A\| = \sum_{i=1}^n \sum_{j=1}^n |a_{ij}|$$

Show that this is a matrix norm (that is, a norm on the linear space of all $n \times n$ matrices). Show that it is not subordinate to any vector norm. Does it conform to Equation (9) and Inequality (10)?

$$\|I\| = 1 \quad (\text{Equation 9})$$

$$\|AB\| \leq \|A\|\|B\| \quad (\text{Inequality 10})$$

Solution:

Problem 5

Problem Statement (Textbook 4.4.11):

Show that for the vector norm $\|\mathbf{x}\|_1$ defined in Equation (5), the subordinate matrix norm is

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|$$

where

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i| \quad (\text{Equation 5})$$

Solution:

Problem 6

Problem Statement (Textbook 4.4.33):

For any $n \times n$ matrix A , define

$$\|A\|_F = \left(\sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 \right)^{1/2}$$

(This is called the **Frobenius** norm.) Is this a subordinate matrix norm? Answer the same question for $\|A\| = \max_{1 \leq i, j \leq n} |a_{ij}|$. Prove that these equations define norms on the vector space of all $n \times n$ matrices.

Solution:

Problem 7

Problem Statement (Textbook 4.6.1b):

Program the Gauss-Seidel method and test it on the following system:

$$3x + y + z = 5$$

$$3x + y - 5z = -1$$

$$x + 3y - z = 3$$

Analyze what happens when these systems are solved by simple Gaussian elimination without pivoting.

Solution:

Problem 8

Problem Statement (Textbook 6.1.1d):

Find the polynomial of least degree that interpolates these sets of data:

x	3	7	12
y	12	146	2

Solution:

Problem 9

Problem Statement (Textbook 6.1.8):

Discuss the problem of determining a polynomial of degree at most 2 for which $p(0)$, $p(1)$, and $p(\xi)$ are prescribed, ξ being any preassigned point.

Solution:

Appendix

References

- [1] David Kincaid and Ward Cheney. *Numerical Analysis: Mathematics of Scientific Computing*. American Mathematical Society, Providence, RI, 3rd edition, 2009. Available from ProQuest Ebook Central. <https://ebookcentral.proquest.com/lib/wsu/detail.action?docID=4717637>.
- [2] Alexander Panchenko. Math 448 lecture notes. Washington State University, Fall 2025.